Human-Computer Interaction

Johan F. HOORN
Jeff TANG

Course objectives

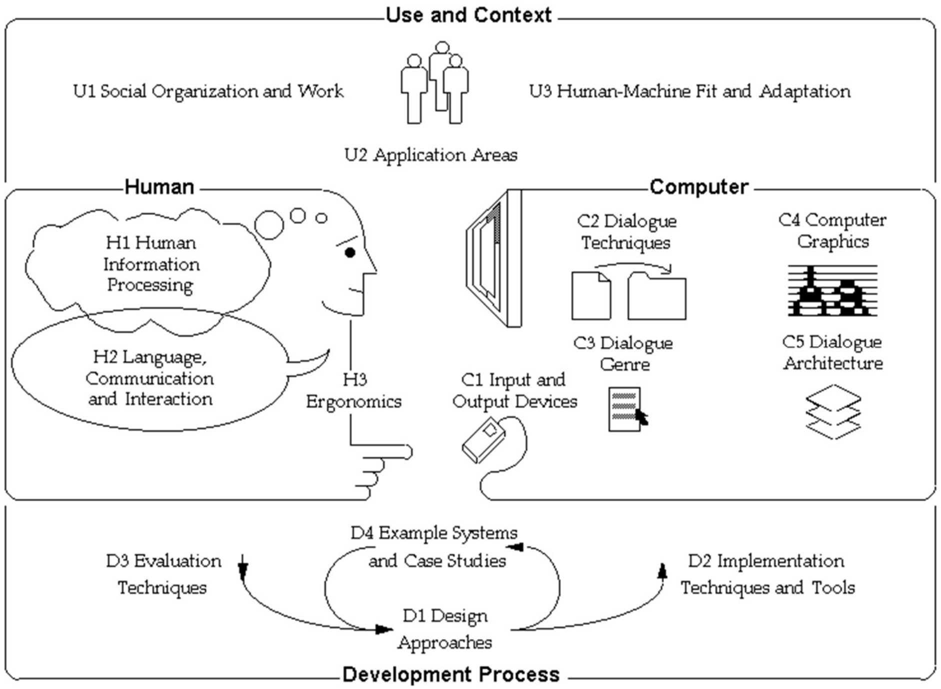
By the end of the course, you have learned:

- ❑ Methods for grounding your design in reality
- ❑ Methods for prototyping (visual) applications
- ❑ Methods for evaluating interface quality
- ❑ Fundamentals of screen design and representations
- ❑ How to apply guidelines to interface design
- ❑ Have sufficient background to continue your education as an Interaction Designer

Course overview

Lecture topics:

1. Use and context	U1-3
2. Human information processing	H1
3. Visual perception and computer graphics	H1, C4
4. Language, communication, and dialogue techniques	H2, C2,3,5
5. Ergonomics, I/O devices, haptics, sound	H3, C1
~~Design approaches and implementation~~	~~D1-2~~
6. Evaluation techniques and example systems	D3-4

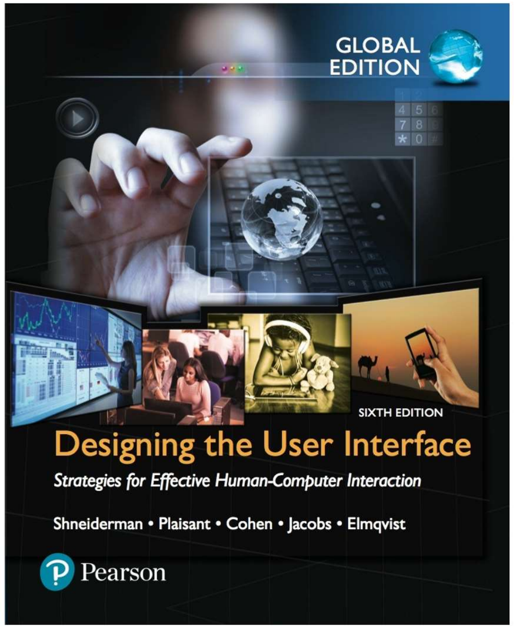


http://www.acm.org/sigchi/cdg/

Model of HCI
SIGCHI (1992)

How we use the book

- ❑ Slides are leading!
- ❑ The book is for background information only
- ❑ We will not discuss chapters one after the other
- ❑ If you have questions about the book, ask us after class



Reading per week:

1. Afterword: Societal and Individual Impact of User Interfaces, p. 578
   Chapters 1, 2, and 3
2. Chapters 12, 13, and 15
3. Chapter 16
4. Chapters 9, 11, and 14
5. Chapters 7, 8, and 10
6. Chapter 4
7. Chapters 5 and 6

Course rundown COMP3423

N001	Y302	Session	Contents	Teacher
Sep 4,	Sep 5	1	Intro HCI 1	Johan / Jeff tutorial
11,	12	2	Intro HCI 2	
18,	19	3	Human 1	
25,	26	4	Human 2	
Oct X,	Oct 3	5	Visual 1	
9,	10	6	Visual 2	
16,	17	7	Language 1	
X,	24	8	Language 2	
30,	31	9	Ergo_I/O 1	
Nov 6,	Nov 7	10	Ergo_I/O 2	
13,	14	11	Testing 1	
20,	21	12	Testing 2	
27,	28	13	Response, test exam	Johan, Jeff

Oct 2 is holiday after National Day. Come Oct 3 (15h30) to Y302. Oct 23 is Chung Yeung festival. Come Oct 24 (15h30) to Y302.

Tutorials (= 3rd hour of class) start from Sept 11:

Introduction to the assignment	Johan
Coaching, guidance, execution	Jeff + TAs

HCI Methods

Johan F. HOORN and Jeff TANG

Lecture 01: Use and Context

Contents

- ❑ Importance of HCI
- ❑ Shocking examples
- ❑ Definitions
- ❑ History
- ❑ Various disciplines
- ❑ Summary

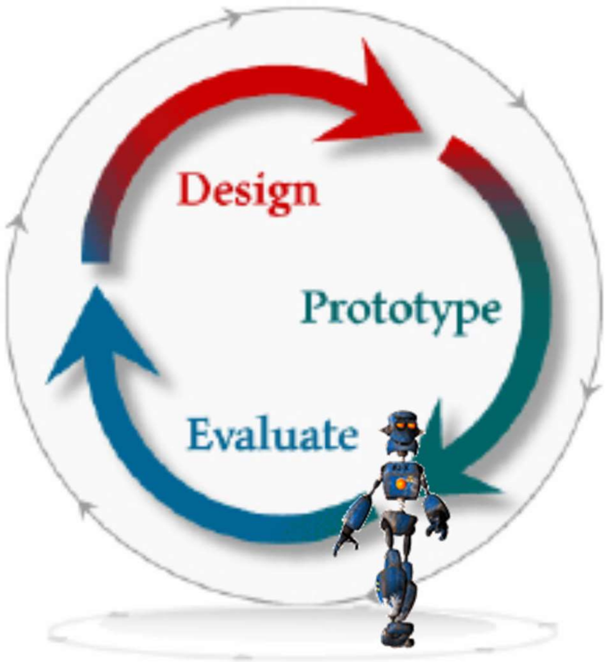
Slides in this lecture are partially based on those from Ms. Tiffany Tang, Dr. Vincent Ng, Dr. Grace Ngai, and Dr. Saul Greenburg

You first quiz item

❑ Where do I come from?

Importance of HCI

Human-Computer Interaction (HCI) 
Computer-Human Interaction (CHI) 
User-Centred Design
Interaction Design



System development

Requirements
Analysis
Design
Code
Test
Implement/deploy

Importance of HCI

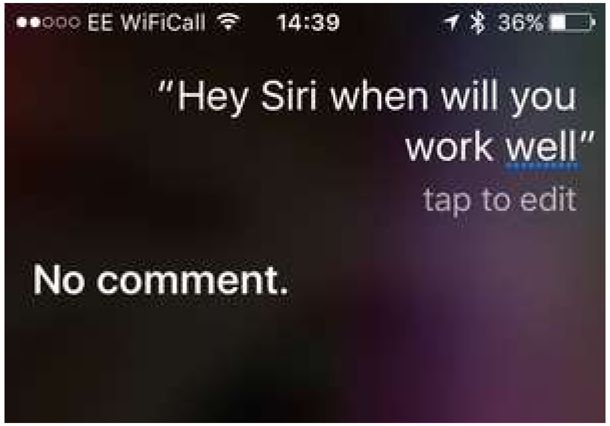
❑ 63% of large software projects go over cost

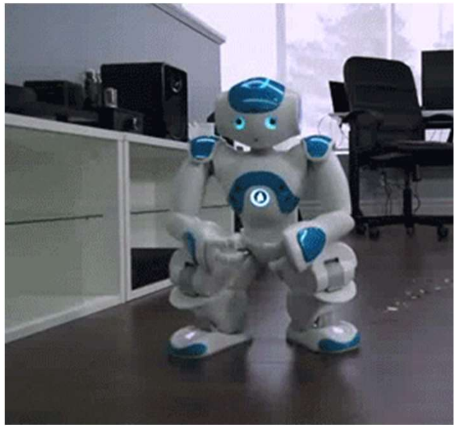
- Managers gave four user-related reasons:
  - Users requested changes
  - Overlooking tasks
  - Users did not understand their own requirements
  - Insufficient user-developer communication and understanding

❑ Usability engineering is software engineering

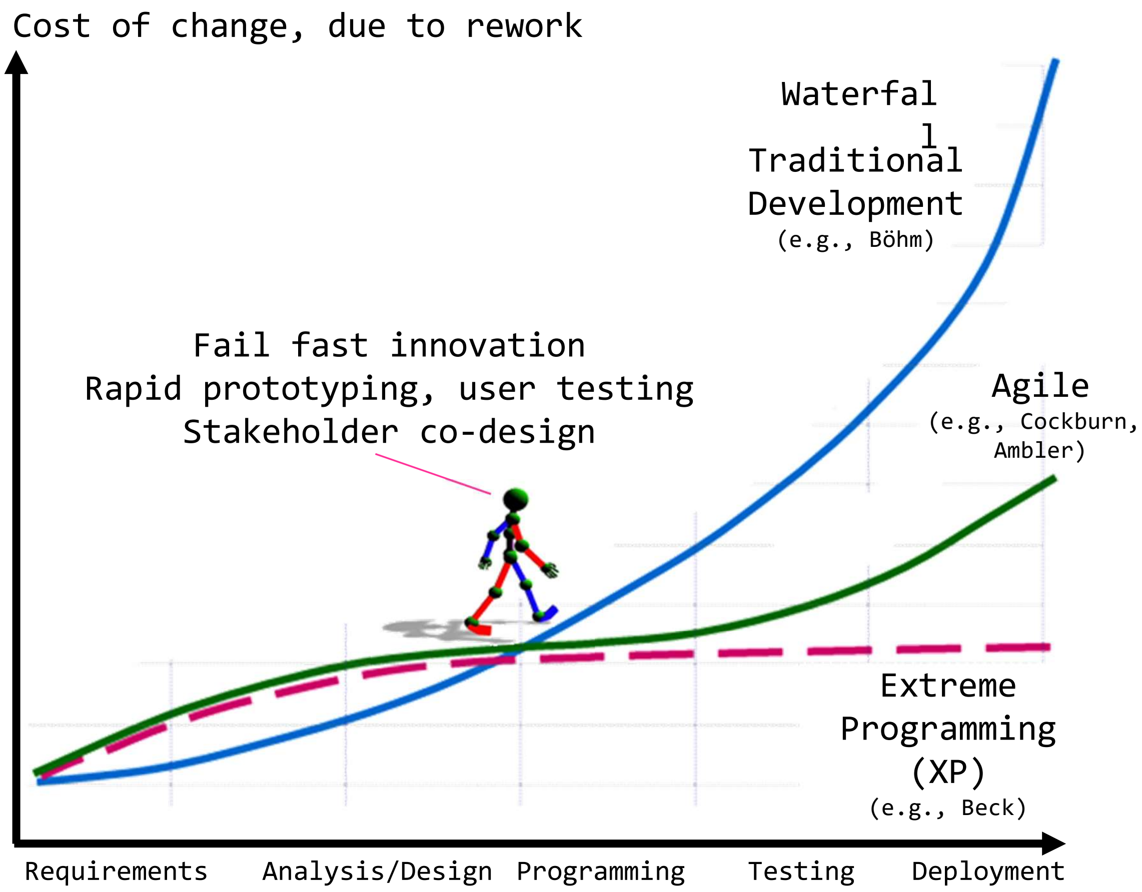
- Coders jump too quickly into detailed design:
  - Founded on incorrect requirements
  - Has inappropriate dialog flow
  - Is not easily used
  - Is never tested until it is too late

- Pay a little now, or pay a lot later!



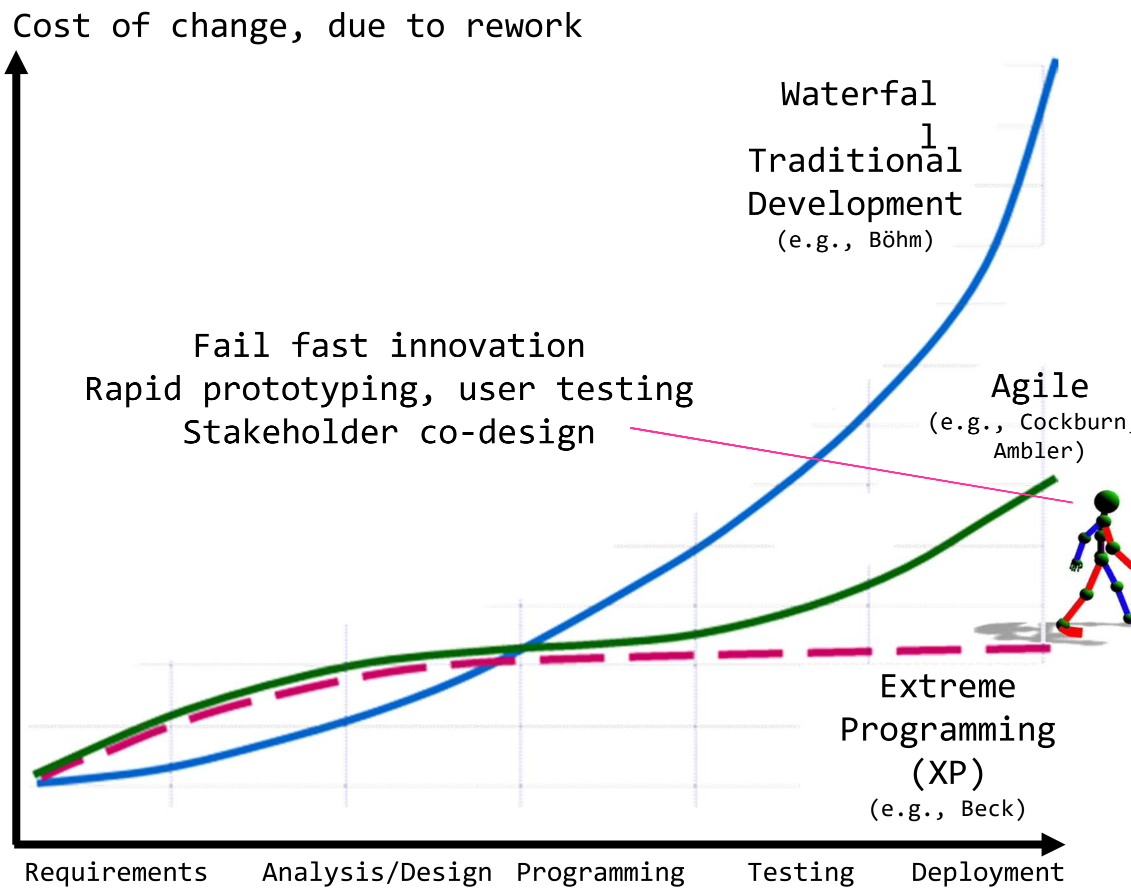


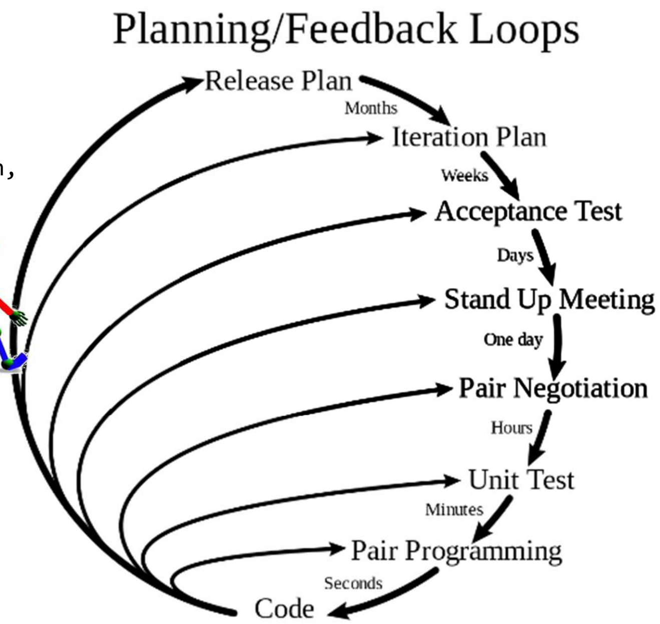
Excellent program-ming job by Prof. Johan

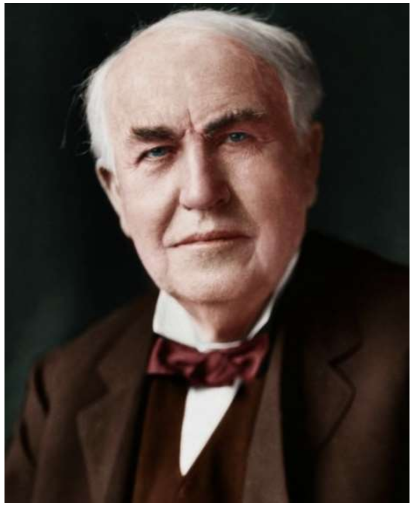


Lous, P., Tell, P., Michelsen, C. B., Dittrich, Y., & Ebdrup, A. (2018). From Scrum to Agile: A journey to tackle the challenges of distributed development in an Agile team. In *Proceedings of the 2018 International Conference on Software and System Process (ICSSP '18)* (pp. 11-20). New York: Association for Computing Machinery. doi: 10.1145/3202710.3203149

Sedano, T., Ralph, P., & Péraire, C. (2016). Sustainable software development through overlapping pair rotation. In *Proceedings of the 10th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement - ESEM '16* (pp. 1–10). doi: 10.1145/2961111.2962590.







Thomas Alva Edison
(1847-1931)

Fail fast innovation
Rapid prototyping, user testing
Stakeholder co-design

Traditional R&D
Waterfall
Agile software development

"I have not failed 1,000 times. I have not failed once. I have succeeded in proving that those 1,000 ways will not work. When I have eliminated the ways that will not work, I will find the way that will work."

The system is redesigned many times before implementation

Low
(continuous improvement and adaptation before deliverance)

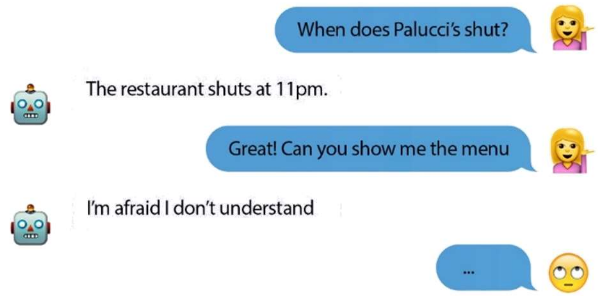
Financial risk
Time expenditure

High
(everything should be right at first deliverance)

Redesign of an implemented system is costly

❑ Today: Usability sells

- Product reviews emphasize usability (e.g., Consumer Reports)
- “Interface has become an important element in garnering good reviews” (Telles, 1990)
- Customers have used related products, and can often download trial versions (including competitors’)
- Less emphasis on features: Today’s users are impatient and intolerant of bad design



❑ Consequences of bad design are large

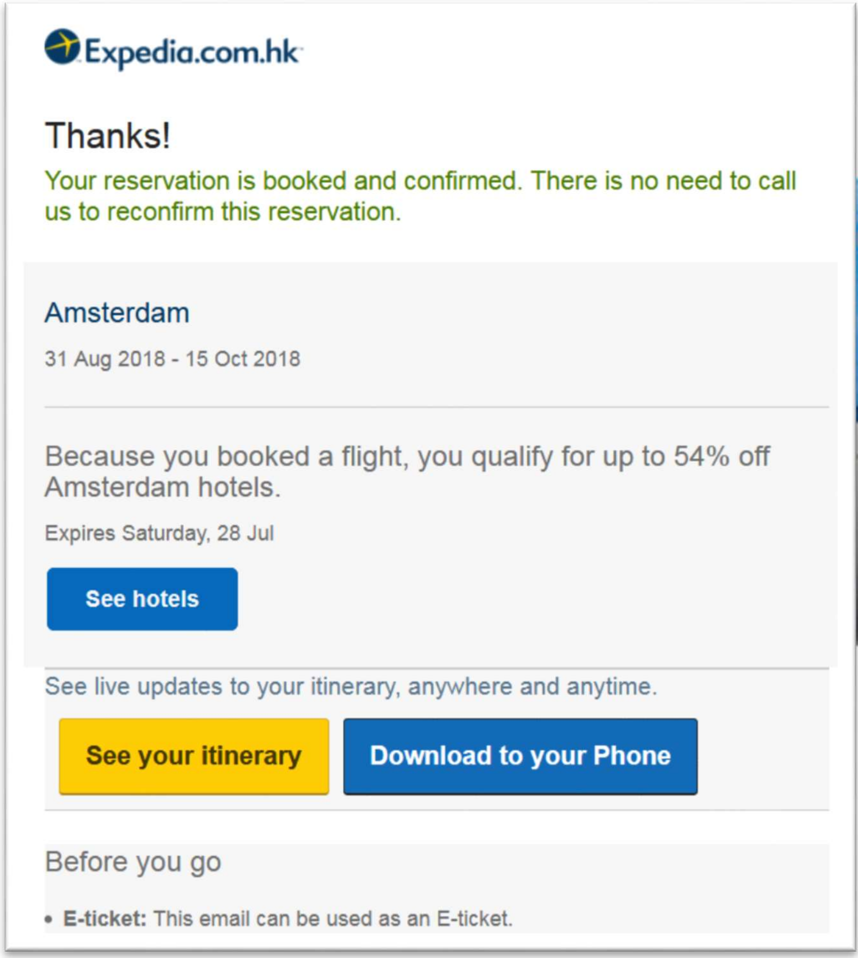
- Costly errors in serious systems (e.g., financial institutes)
- Widespread effects (e.g., incorrect billing, failures)
- Life-critical systems (medical, air traffic control)
- Safety and security (in-car navigation systems, face-ID hacks)


abc 15 ARIZONA
SELF-DRIVING UBER KILLS WOMAN
abc 15 ARIZONA

Daily life example

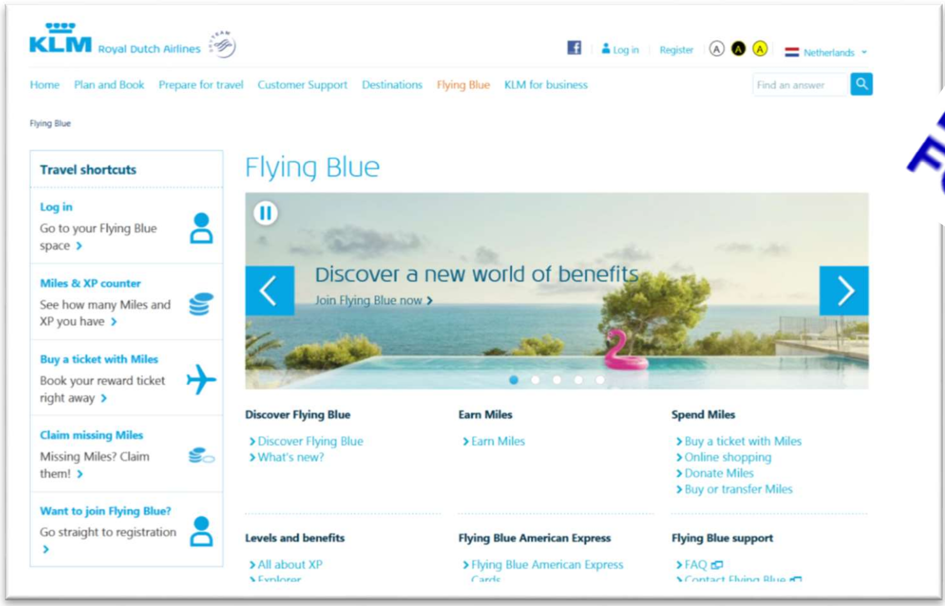
August 2018

I book and pay a KLM flight with Expedia, economy class



Daily life example

I want to upgrade with my miles collected with KLM
I upgrade my seat at KLM website
But in the website there is no option to pay with miles



I abort the session, assuming that my old reservation is still valid (the upgrade was not paid)
I write KLM why I can’t upgrade with their own miles
KLM explains why this is impossible (whatever)

Daily life example

Thirty hours before boarding, I check in online.
Turns out, my entire reservation is cancelled

I call the airliner KLM, they repair the cancellation on site
Luckily, there was 1 chair left (otherwise, I had to go the next day)

Time and money wasted. How could it go wrong like this?

Re: Expedia travel confirmation - 31 Aug - (Itinerary # 7364492520082)

Johan Hoorn<jf.hoorn@gmail.com> Thu, Aug 30, 2018 at 9:31 AM
To: Expedia@expediamail.com

Reply | Reply to all | Forward | Print | Delete | Show original

Dear Expedia,

I see my reservation was canceled but that is not true. I tried to upgrade my ticket with Flying Blue but that did not work. I never cancelled anything. I have to be on that flight. Please help!

Johan Hoorn

On Sun, 8 Jul 2018, 10:13 Expedia, <Expedia@expediamail.com> wrote:

Expedia.com.hk

Daily life example

KLM explains that Expedia cancelled the reservation, because I bought the ticket with them

KLM cannot handle tickets sold by an agent

What is wrong with the HCI design of this system?

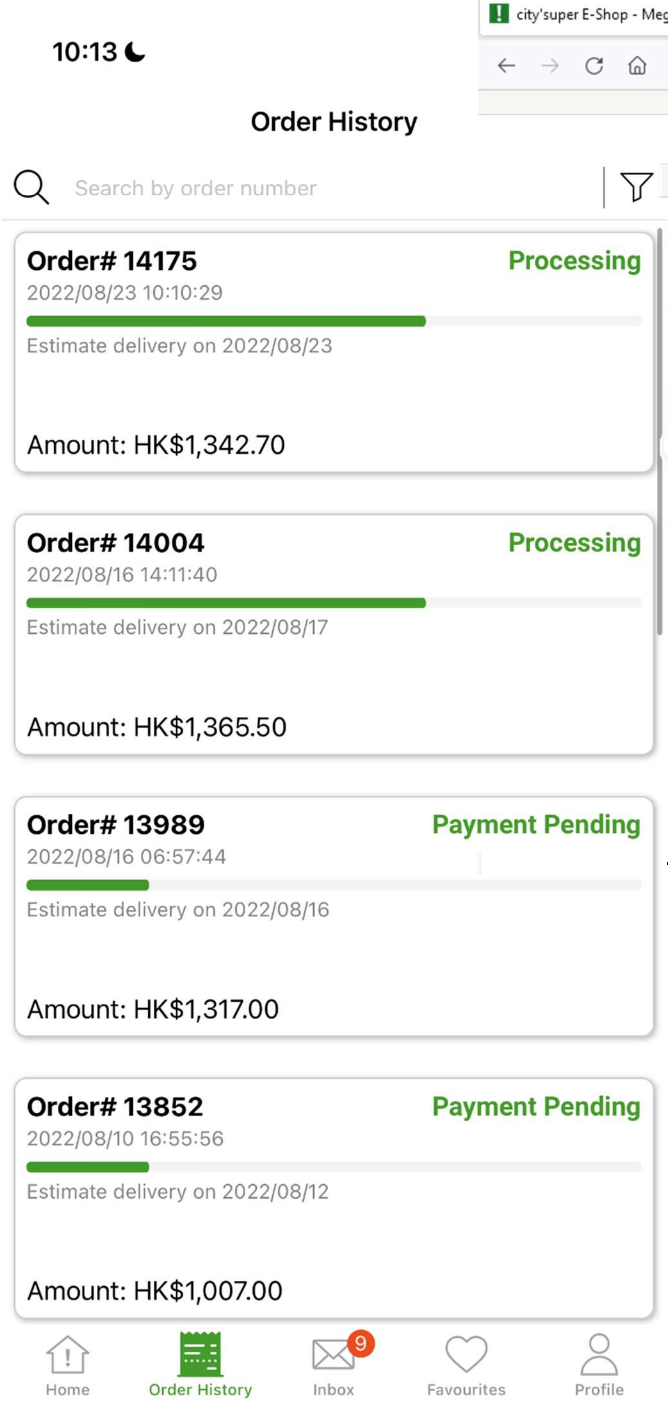
- Flying Blue (the air-mile system) should state that you cannot upgrade with miles
- KLM website should check with agencies where their tickets are booked (and warn users that want impossible things)
- Expedia should inform its clients on changes in reservation status
- That KLM cannot handle Expedia tickets is part of a business process, which should not bother your customer

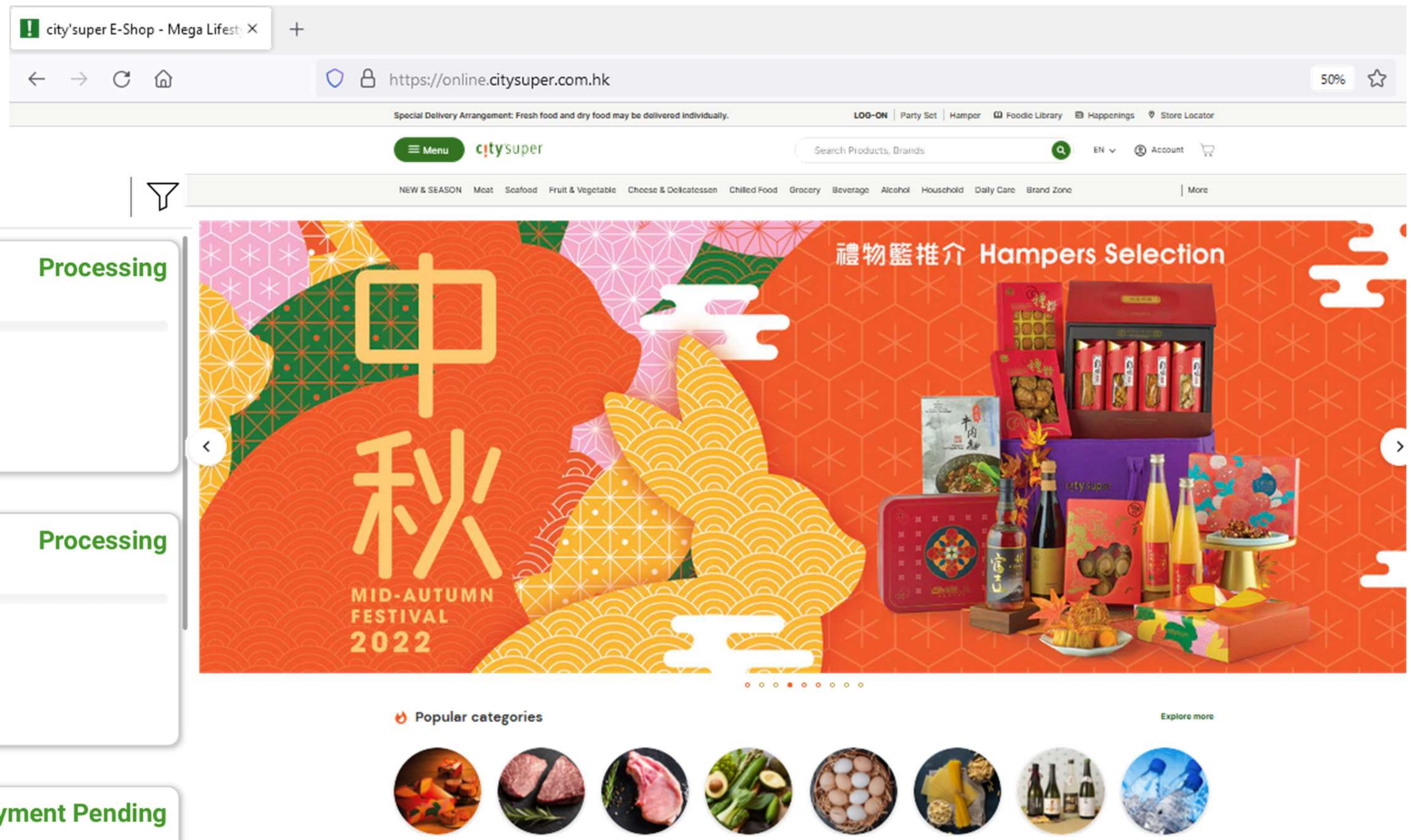
Importance of HCI

Given this example, is HCI just about interface design?

No, HCI also is concerned with the customer journey, organization of work, task delegation, business processes, the algorithms; anything to improve the user experience

- Flying Blue (the air-mile system) should state that you cannot upgrade with miles
- KLM website should check with agencies where their tickets are booked (and warn users that want impossible things)
- Expedia should inform its clients on changes in reservation status
- That KLM cannot handle Expedia tickets is part of a business process, which should not bother your customer





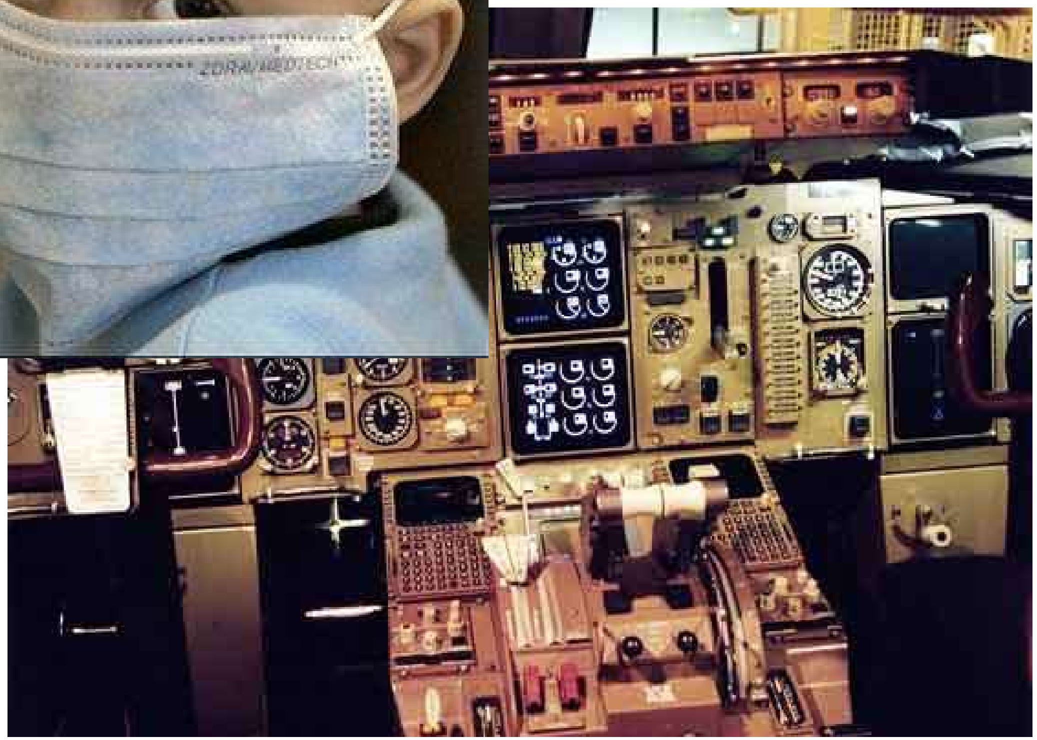
- ❑ Parcels delivered but still processing
- ❑ What goes wrong here? What are the consequences?
- ❑ Congestion in data network causes low service quality, queueing delay, packet loss, blocking new connections
- ❑ Usability engineering = software engineering

HCI Methods

Johan F. HOORN and Jeff TANG

Lecture 01: Use and Context

Shocking examples



Fat finger costs Japanese broker US$225m

(Guardian/Agencies)
Updated: 2005-12-10 06:34

So-called fat finger errors, while rare in electronic financial markets, are accepted as an inevitable risk when humans come into contact with computers in frenetic trading conditions. But such mistakes are usually spotted quickly, with the resultant "rogue" trades being unwound between the affected parties.

An unnamed and, presumably, shortly to be unemployed broker, managed to sell 610,000 shares at 1 yen (less than a penny) apiece in a job recruiting firm called J-Com Co, which was having its public debut on the exchange. It had actually intended to sell 1 share at 610,000 yen (US$5,041).

Problem:

- Fat fingers
- Information overload
- Human error
- Inflexible transaction system

Mizuho Securities President Makoto Fukuda (foreground) bows in apology at a news conference on Thursday, after the brokerage made a huge errant order in newly listed shares of small recruitment firm J-Com Co. [Reuters]

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Fat finger costs Japanese broker US$225m

(Guardian/Agencies)
Updated: 2005-12-10 06:34

In the Mizuho case, however, no immediate action seems to have been taken, leaving scores, if not hundreds, of investors across the Tokyo market to carry on trading stock that had been sold in error.

Mizuho says it tried to cancel the order three times, but the exchange said it does not cancel transactions even if they are executed on erroneous orders.

Problem:

- ❑ Fat fingers
- ❑ Information overload
- ❑ Human error
- ❑ Inflexible transaction system

Mizuho Securities President Makoto Fukuda (foreground) bows in apology at a news conference on Thursday, after the brokerage made a huge errant order in newly listed shares of small recruitment firm J-Com Co. [Reuters]

Importance of HCI

- Fat fingers
- Information overload
- Human error
- Inflexible transaction system

- Ergonomic design
- Reduce short-term memory load
- Offer error prevention
- Permit easy reversal of actions

Design a solution: How would you prevent selling 610,000 shares at 1 yen apiece instead of 1 share at 610,000 yen?

If you don't, here is what happened:

- Confidence in Tokyo Stock Exchange damaged
- Shares in NTT, rival of software maker, jumped 11%
- $331,000,000 loss from trade for Mizuho

Let's bow our heads for button makers

University of Bielefeld - Faculty of technology

Networks and distributed Systems

Research group of Prof. Peter B. Ladkin, Ph.D.

http://sunnyday.mit.edu/accidents/calirep.html

AA965 Cali Accident Report

Near Buga, Colombia, Dec 20, 1995

Prepared for the WWW by
Peter Ladkin
Universität Bielefeld
Germany

Prepared November 6, 1996

AERONAUTICA CIVIL
of THE REPUBLIC OF COLOMBIA

SANTAFE DE BOGOTA, D.C. - COLOMBIA

AIRCRAFT ACCIDENT REPORT

CONTROLLED FLIGHT INTO TERRAIN
AMERICAN AIRLINES FLIGHT 965
BOEING 757-223, N651AA
NEAR CALI, COLOMBIA
DECEMBER 20, 1995

Searleman, 2007

- Pilots manually navigate only during takeoff, landing, bad weather, emergencies
- Navigate by traveling between beacons
- Flight management system

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of THE REPUBLIC OF COLOMBIA

SANTAFE DE BOGOTA, D.C. - COLOMBIA

AIRCRAFT ACCIDENT REPORT

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NEAR CALI, COLOMBIA
DECEMBER 20, 1995

AmericanAirlines

Searleman, 2007

Push **R** to navigate to **ROZO** beacon

But plane navigated to **ROMEO** beacon

No feedback on chosen beacon

Plane headed in the wrong direction, eventually crashing on a mountain

ROZO

Speed brakes not disengaged when trying to accelerate

ROMEO

Searleman, 2007

Mercedes Johnson survived the American Airlines Flight 965, which crashed into a Colombian mountain in 1995. Out of 159 onboard, only four lived

http://news.bbc.co.uk/2/low/uk_news/magazine/5402342.stm

Official cause: “pilot error”

Actual cause: poorly designed HCI

Let’s bow our heads for button makers

http://www.pilotfriend.com/disasters/crash/images/29.jpg

Better known as the Harrisburg meltdown

http://images.jupiterimages.com/common/detail/81/36/23263681.jpg


http://www.velan.com/solutions/media/inservice/nuclear.jpg

- A light indicated a valve had been closed when in fact it hadn’t

Robert Amor

Software Engineering The University of Auckland

http://images.jupiterimages.com/

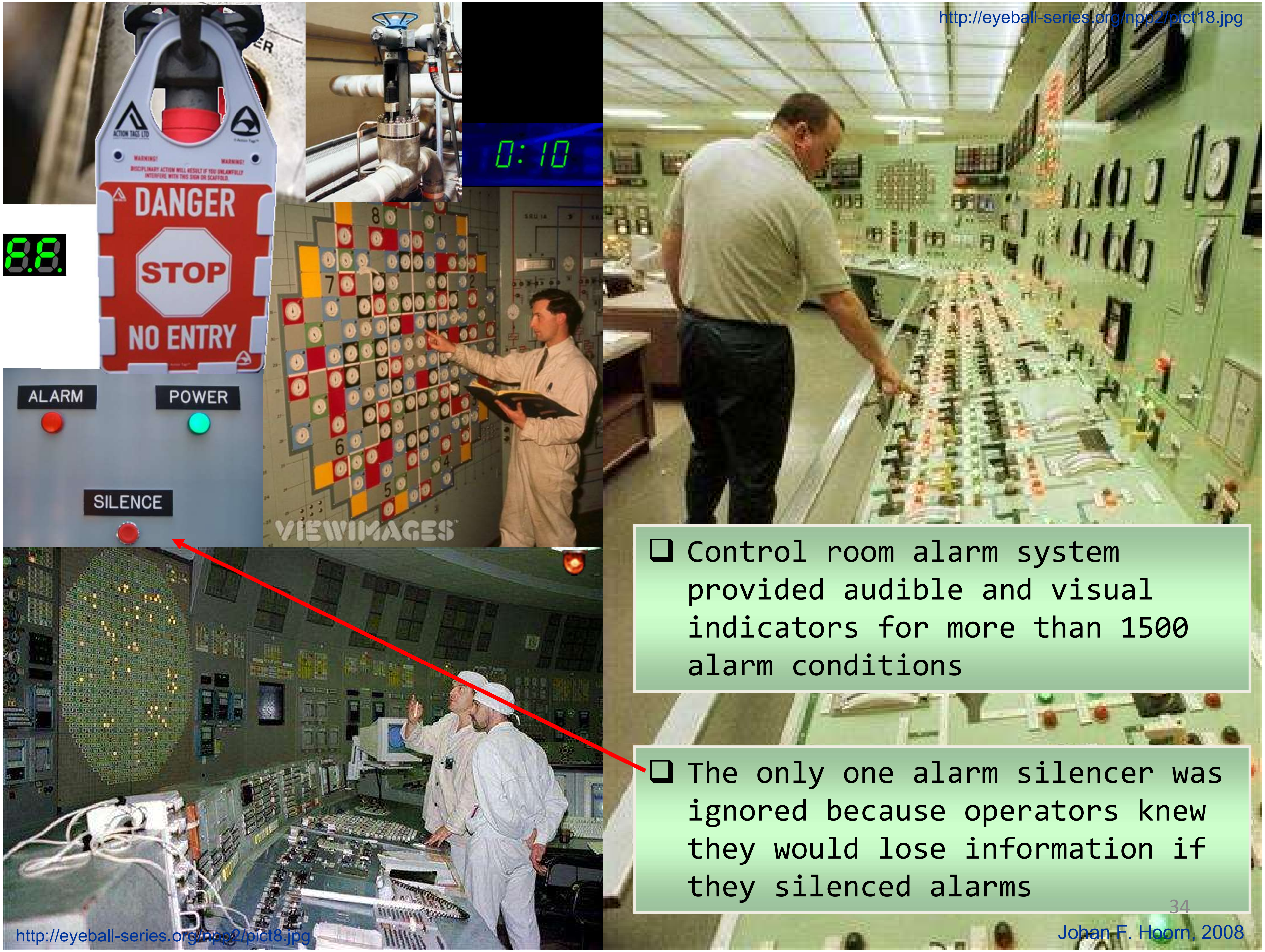
http://www.actiontags.co.nz/file/gallery/large/95_red_danger_no_entry.jpg

http://www.velan.com/solutions/media/inservice/nuclear.jpg

- A light indicated a valve had been closed when in fact it hadn't
- The light indicator was obscured by a caution tag attached to another valve

Robert Amor

Software Engineering
The University of Auckland



- Control room alarm system provided audible and visual indicators for more than 1500 alarm conditions
- The only one alarm silencer was ignored because operators knew they would lose information if they silenced alarms

http://neatorama.cachefly.net/images/2007-02/three-mile-island.jpg

http://inventorspot.com/files/images/cancer%20patient.JPG

Importance of HCI

- ❑ Malfunction of indicator
  (false positive)
- ❑ Information overload
  (1500 alarm indicators)
- ❑ Human error
  (tag in front of indicator)
- ❑ Inflexible alarm system
  (either On or Off - silencer)

- ➢ Offer informative feedback
- ➢ Reduce short-term memory load
- ➢ Offer error prevention
- ➢ Support internal locus of control

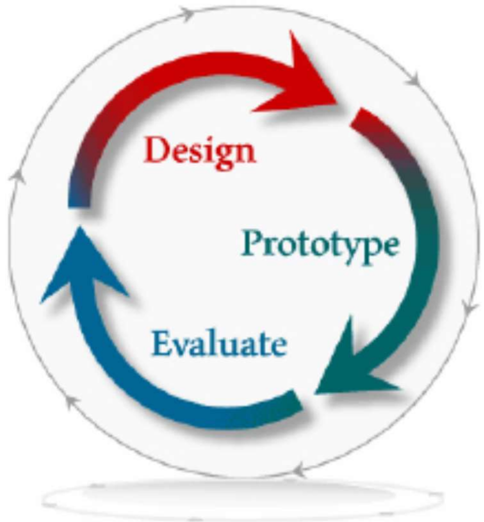
Disaster could have been prevented with better designed control panels

Robert Amor

SE Software Engineering
The University of Auckland

Definitions

- "HCI is… concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them." (ACM SIGCHI Curricula for Human-Computer Interaction)
- "An input language for the user, an output language for the machine, and a protocol for interaction" (CHI 1985)
- "It's simply the parts of the computer that you see, hear, touch or talk to. It is the set of all the things that allow you and your computer to communicate with each other." (IBM Design Concepts)
- A discipline concerned with the design, implementation, and evaluation of interactive computing systems for human use



HCI Methods

Johan F. HOORN and Jeff TANG

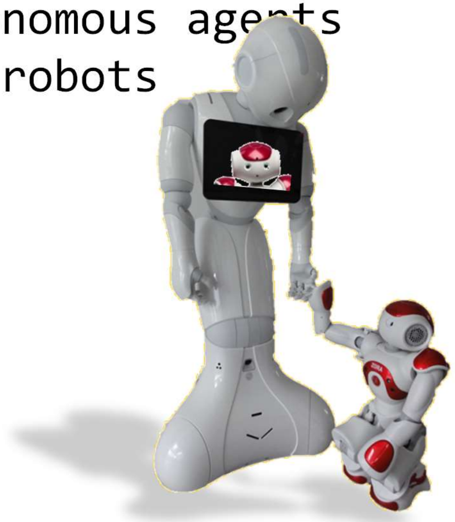
Lecture 01: Use and Context

History of HCI

History of HCI

- ❑ Input/output devices

	Input	Output
Before	Connecting wires	Lights on a display
	Paper tape, punch cards	Paper
	Keyboard	Teletype/teleprinter
Today	Keyboard	Scrolling glass teletype
	+ cursor keys	Character terminal
	+ mouse	Bit-mapped screen
Future	Data gloves + suits	Head-mounted displays
is near	Computer jewelry	Ubiquitous computing
	Natural Language	Autonomous agents
		and robots



- ❑ Keyboards & terminals are artifacts of today's technologies!
- ❑ New I/O devices will change the way we interact with computers!

History of HCI

The way people interacted with computers

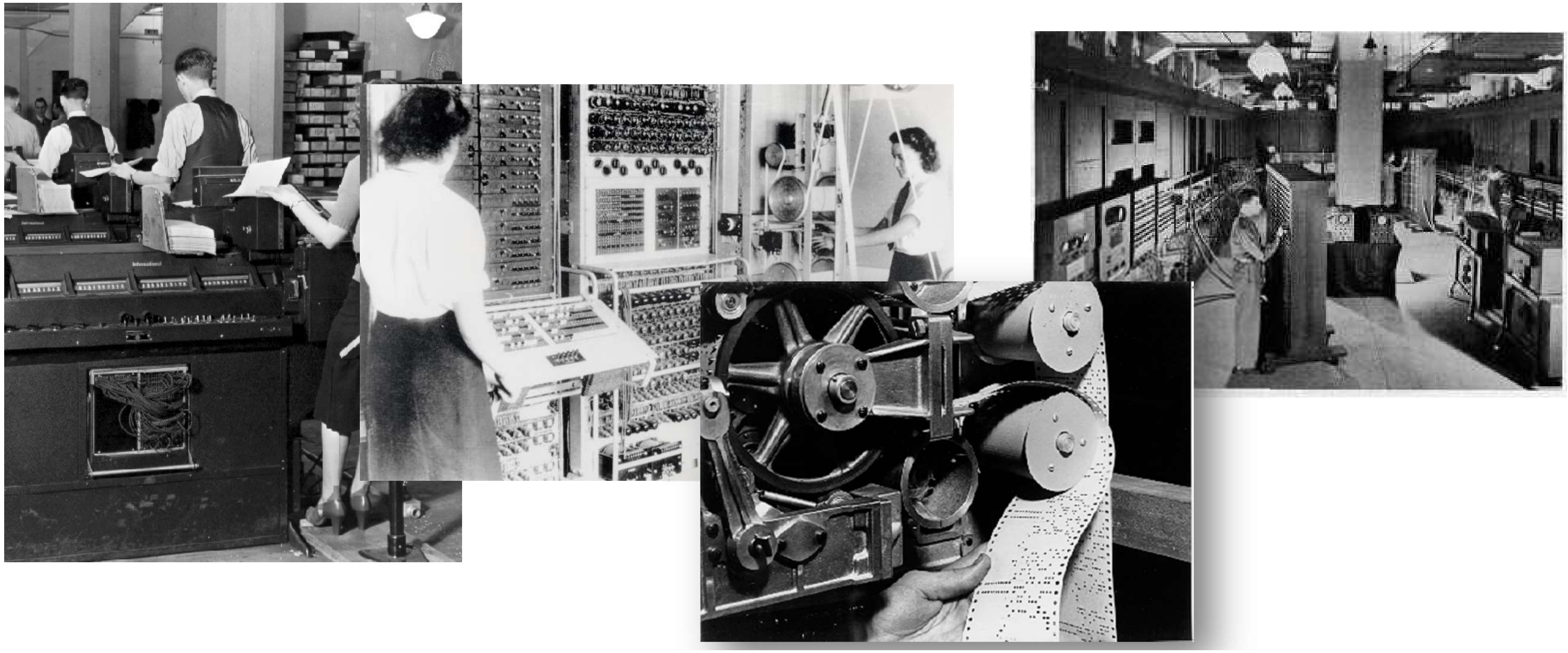
❑ Batch processing	Requests processed group-wise No interaction	Impersonal computing
❑ Timesharing	All work together on the same terminal to save costs	Interactive computing
❑ Networking	Server connects multiple clients for reading and writing data	Community computing
❑ Graphical displays		Direct manipulation
❑ Microprocessors		Personal computing
❑ WWW		Global information
❑ Ubiquitous computing Internet of Things		Embedded software in objects to service everyday activities
❑ Social robots		Companions and coaches for life

History of HCI

A number of examples

Questions to you:

- ❑ What is the role of the user?
- ❑ What does the interaction look like?
- ❑ What is the relation between human and machine?



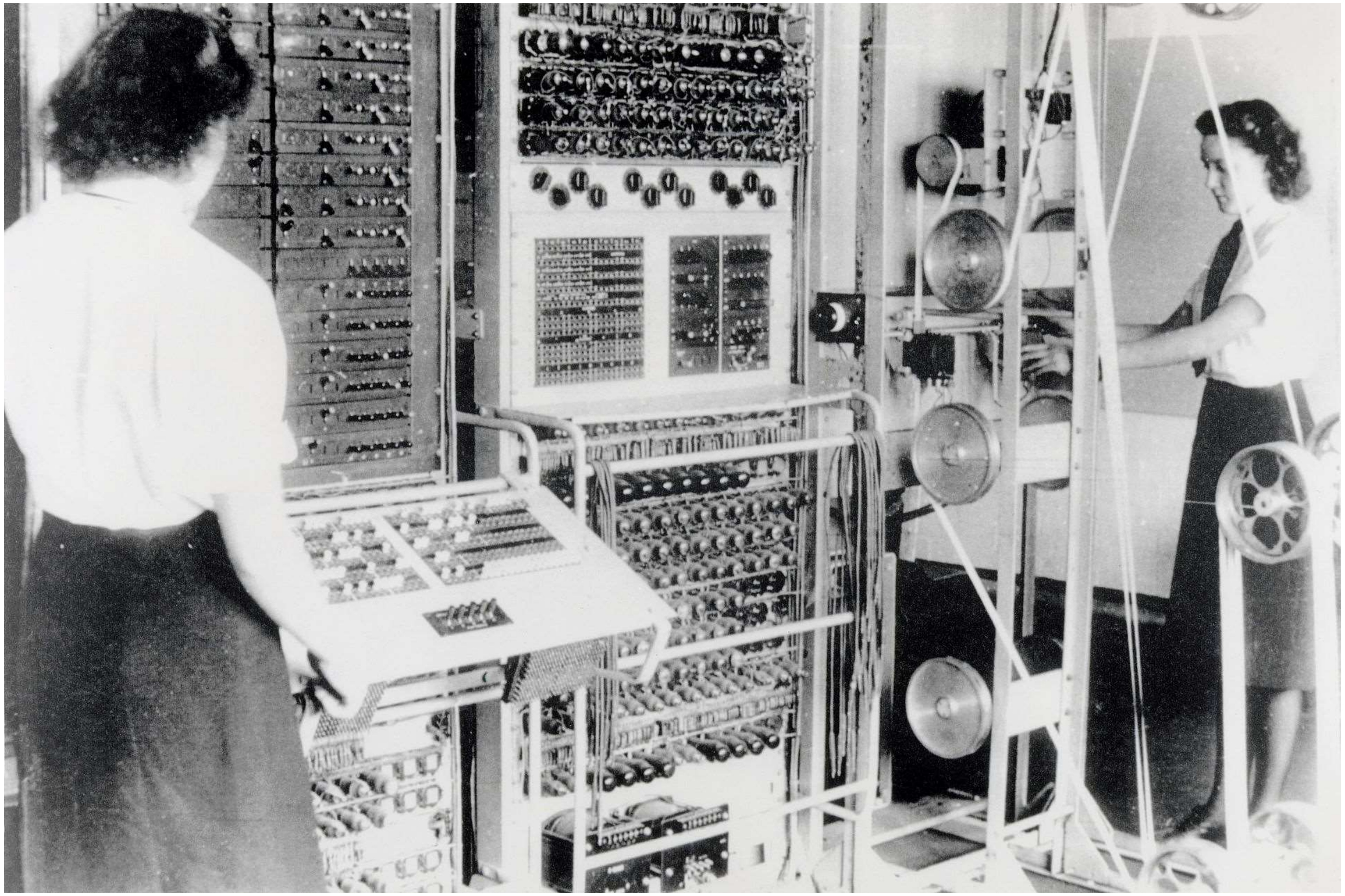
Batch processing

IBM Type 285 tabulators (1936) for batch processing of punch cards in a stack on each machine

United States Social Security Administration

COLOSSUS (1943)

The world's first all electronic numerical integrator and computer

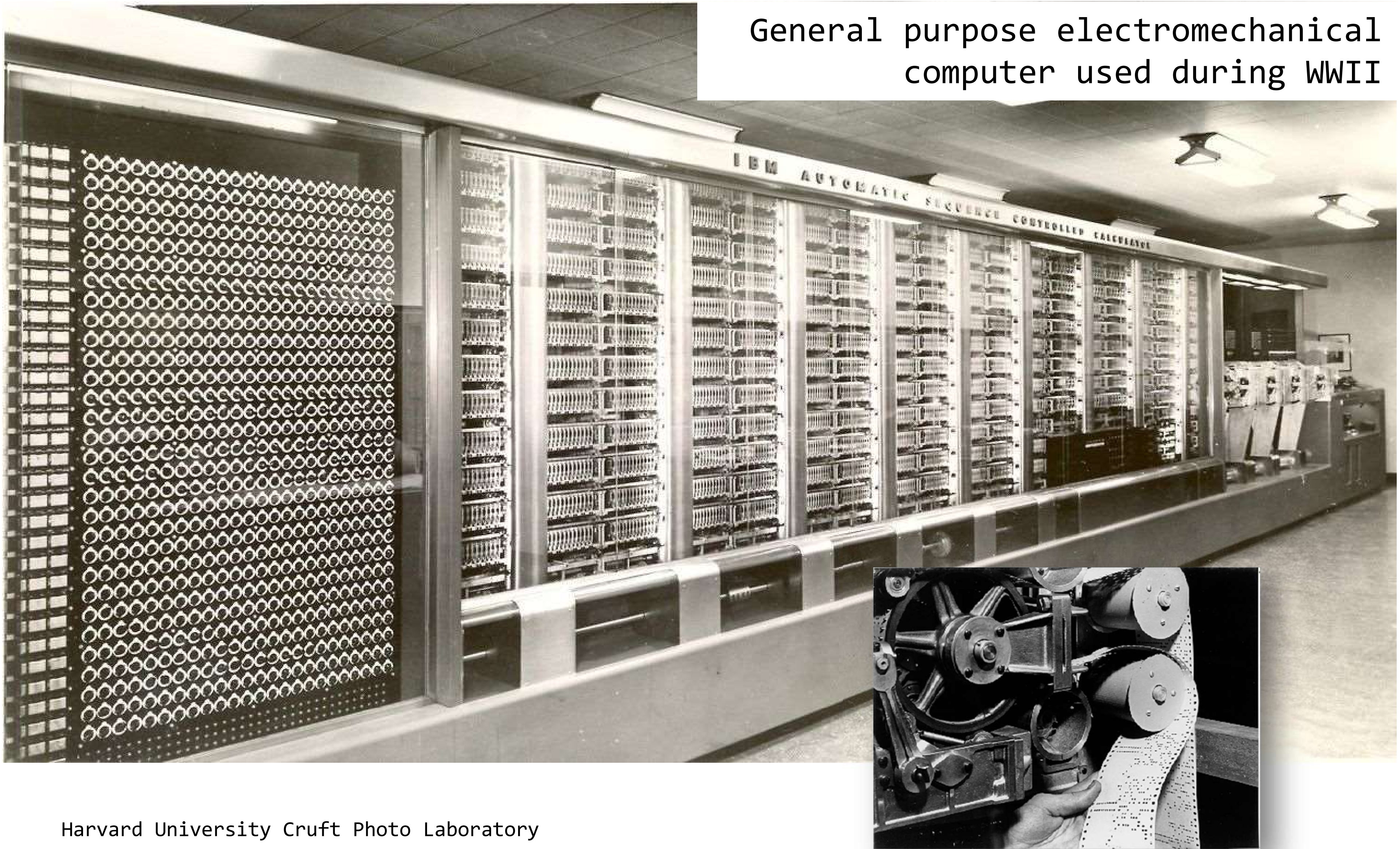


1,500 radio tubes for counters, registers, and logic operations

Punch tape similar to telex machines

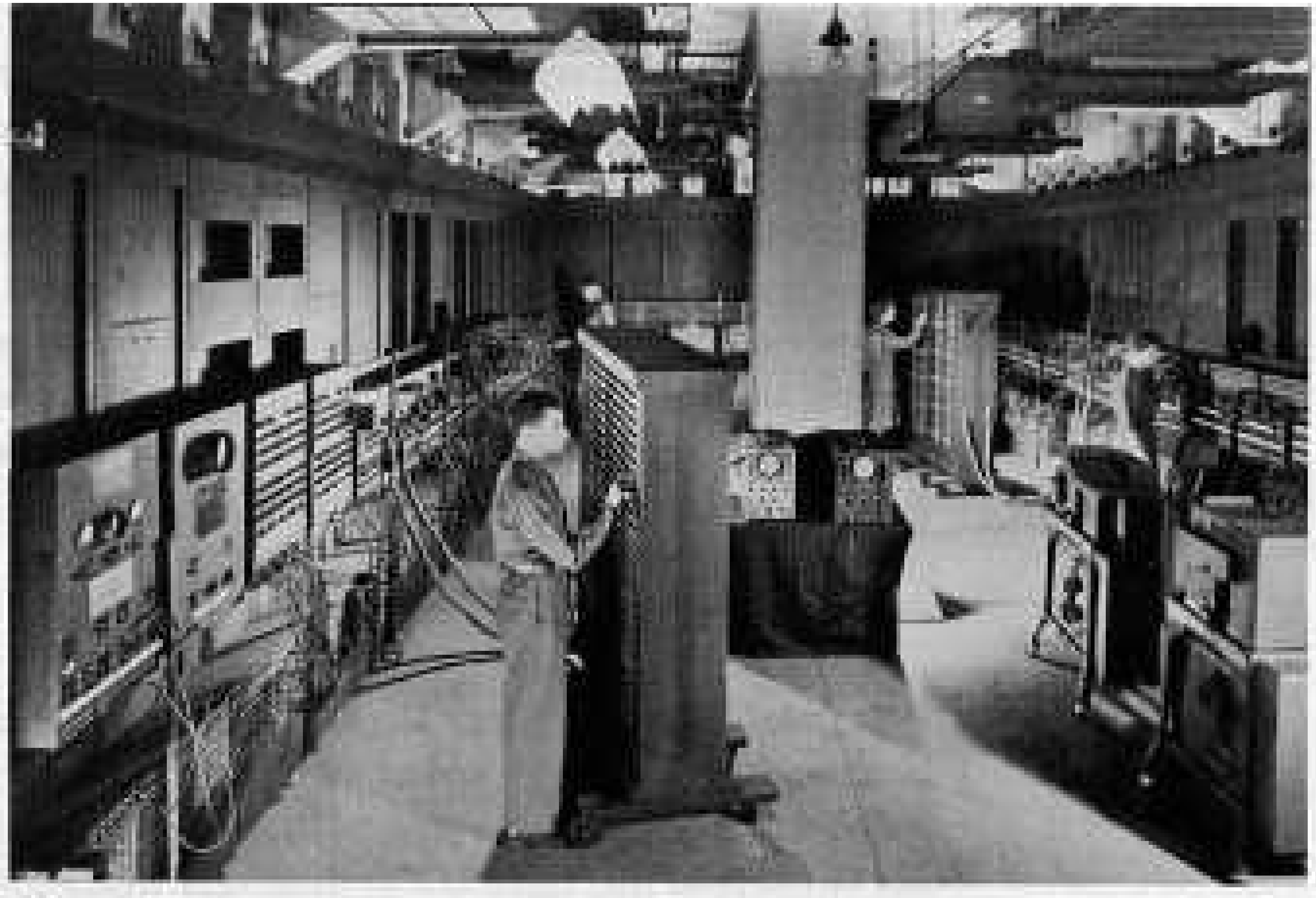
IBM Automatic Sequence Controlled
Calculator (ASCC), called Mark I (1944)

General purpose electromechanical
computer used during WWII



Harvard University Cruft Photo Laboratory

ENIAC (1946)
The world's second all electronic numerical integrator and computer

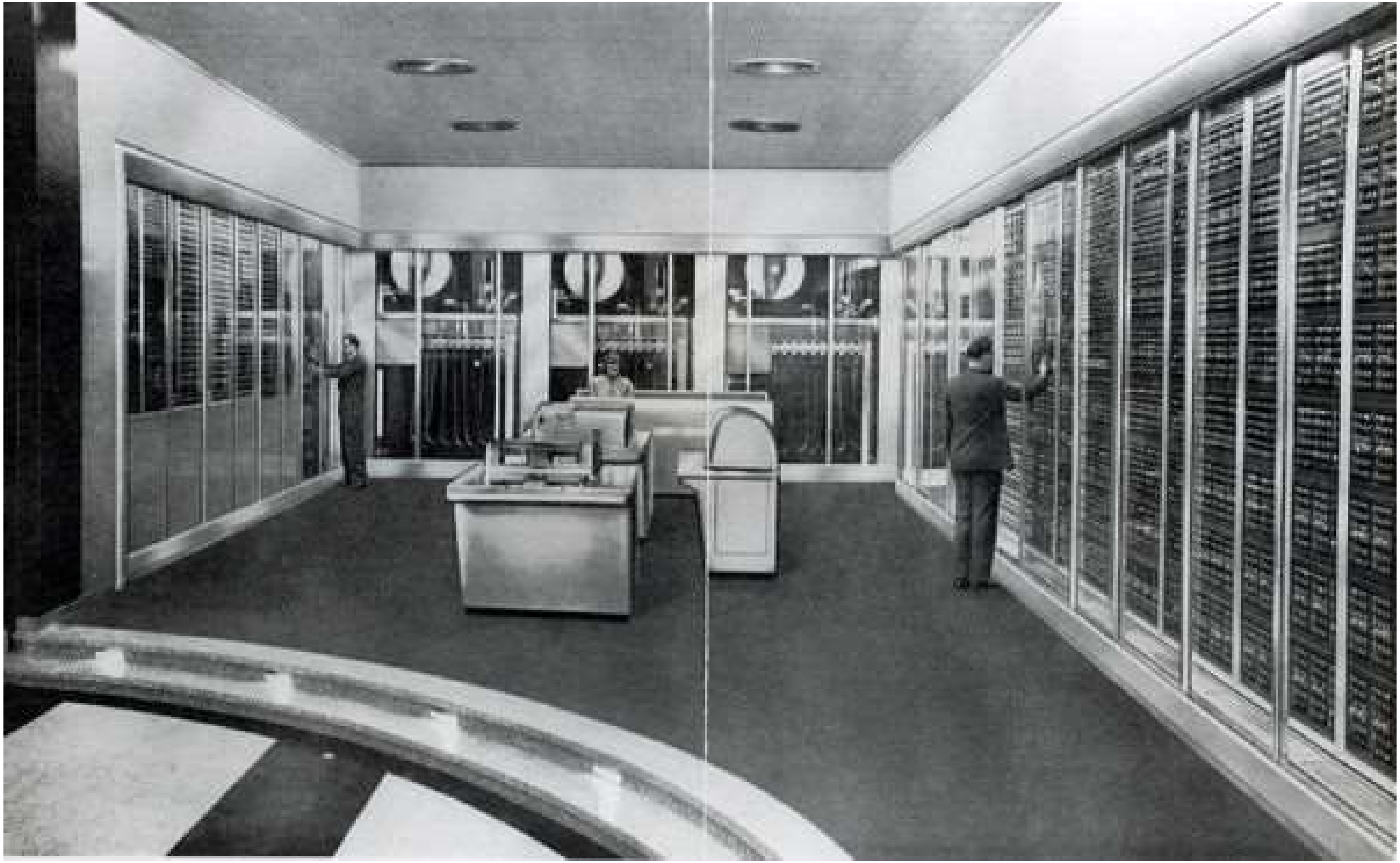


19,000 radio tubes!

IBM Archives

Radio tubes > transistors > microprocessors

IBM SSEC: Selective Sequence Electronic Calculator (1948) - electromechanical computer

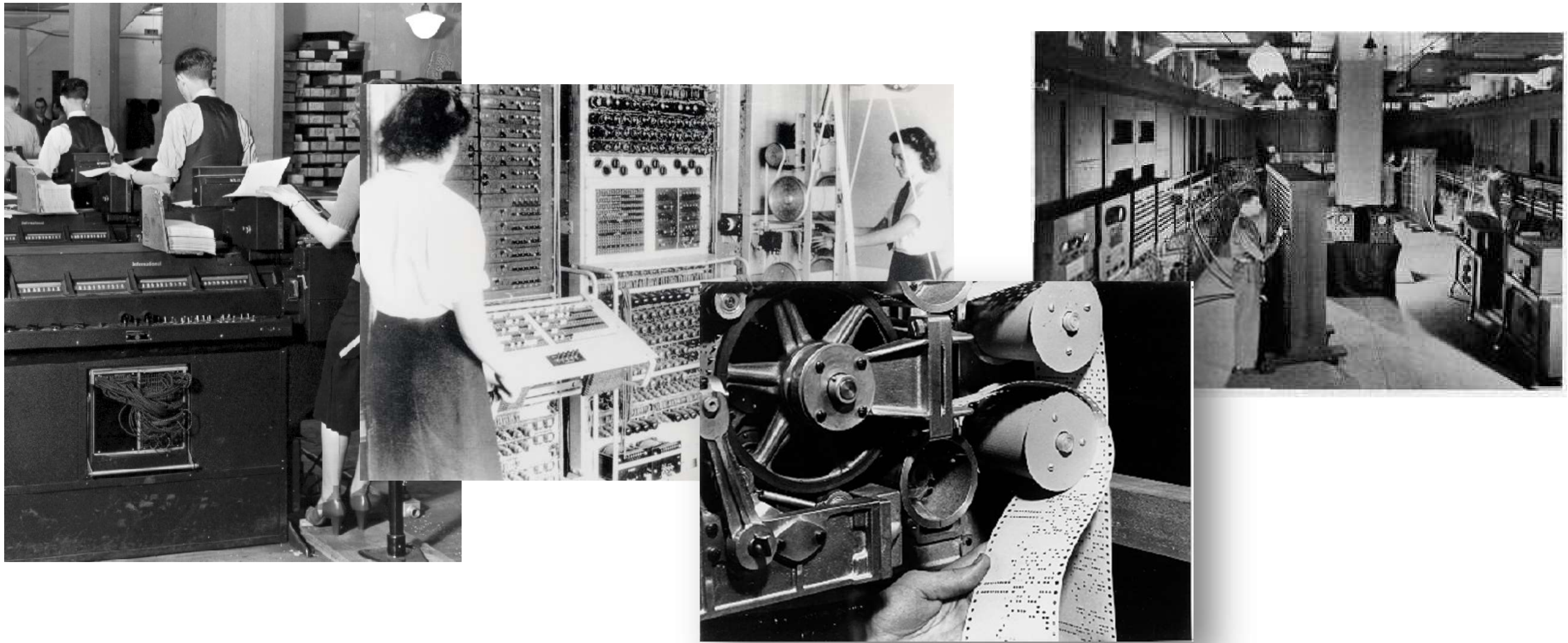


History of HCI

IBM 285 tabulator, COLOSSUS, Mark I, ENIAC, SSEC

Your answers, please:

- ❑ The role of the user is …
- ❑ The interaction is …
- ❑ The relation between human and machine is …

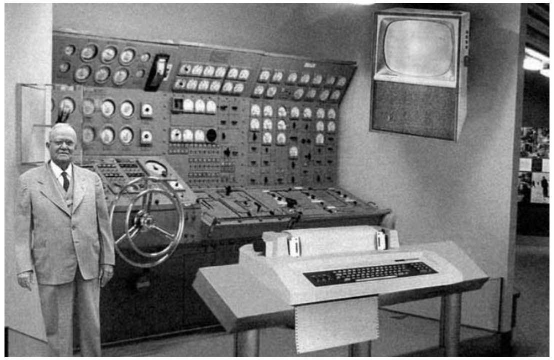


History of HCI

Next batch of examples

Questions to you:

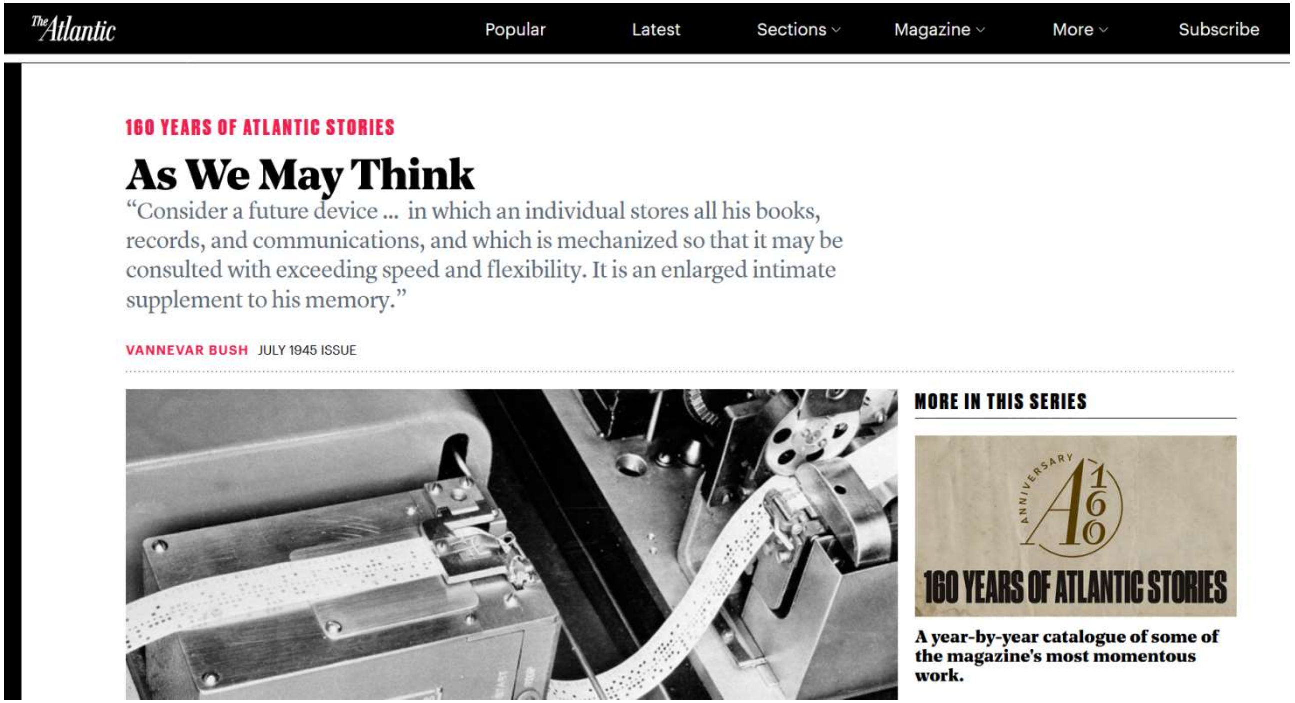
- ❑ What is the role of the user?
- ❑ What does the interaction look like?
- ❑ What is the relation between human and machine?



History of HCI

Intellectual foundations laid by
Vannevar Bush (1945)

- ❑ "As we may think" article in Atlantic Monthly
- ❑ Identified the information storage and retrieval problem: new knowledge does not reach the people who could benefit from it
- ❑ "publication has been extended far beyond our present ability to make real use of the record"

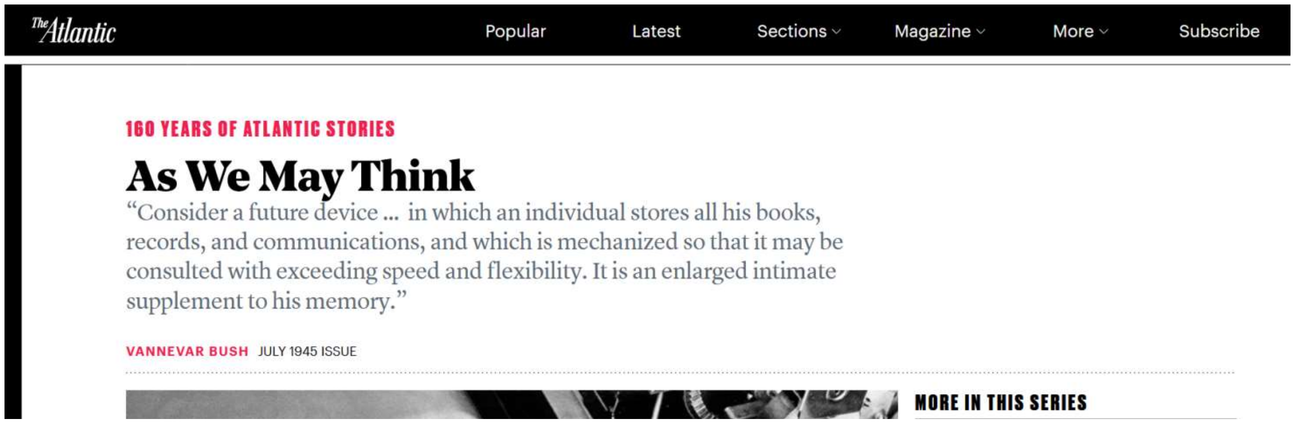


History of HCI

Vannevar Bush’s Memex (Memory Extender) (1945)

Conceiving Hypertext and the World Wide Web

- ❑ Device where individuals store all personal books, records, communications
- ❑ Items retrieved rapidly through indexing, keywords, cross references,...
- ❑ Can annotate text with margin notes, comments...
- ❑ Can construct and save a trail (chain of links) through the material
- ❑ Acts as an external memory
- ❑ Bush’s Memex idea was based on <u>microfilm</u> records but not implemented



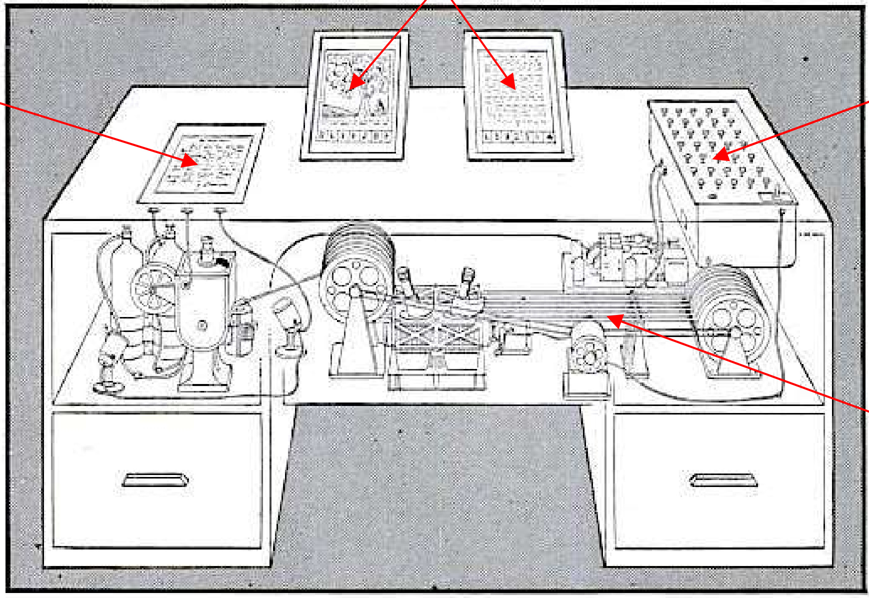
History of HCI

Vannevar Bush’s Memex (1945) (forerunner of WWW)

Two projectors: one for images, one for text (cross references)

Tablet for user input (notes)

User control panel for bookmarks, hyperlinks, and automated search. Content can be uploaded to other Memex machines



Information (text, images) on microfilm

MEMEX in the form of a desk would instantly bring files and material on any subject to the operator's fingertips. Slanting translucent viewing screens magnify supermicrofilm filed by code numbers. At left is a mechanism which automatically photographs longhand notes, pictures and letters, then files them in the desk for future reference.

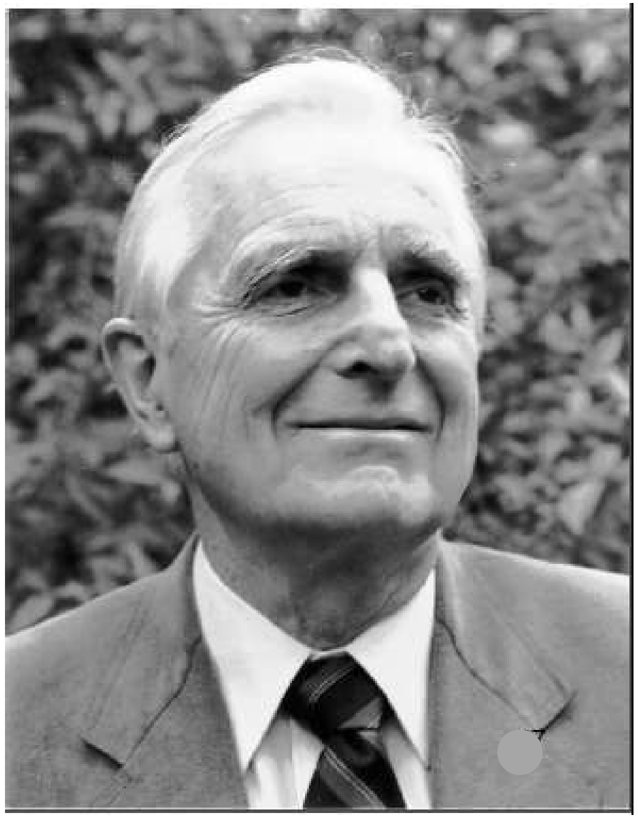
AS WE MAY THINK CONTINUED

History of HCI

Douglas Engelbart (1950s)

“...The world is getting more complex, and problems are getting more urgent. These must be dealt with collectively. However, human abilities to deal collectively with complex / urgent problems are not increasing as fast as these problems.

If you could do something to improve human capability to deal with these problems, then you'd really contribute something basic.”

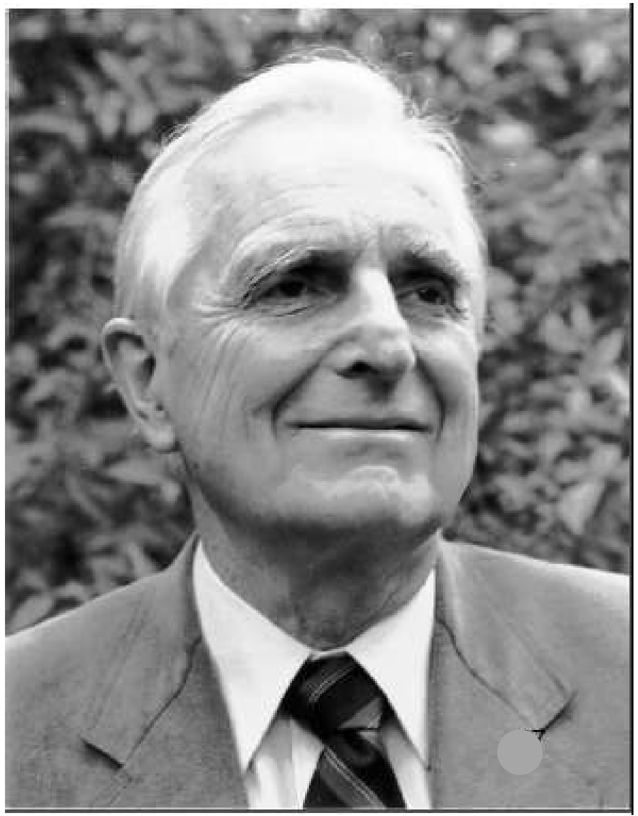


History of HCI

Douglas Engelbart (1950s)

... I had the image of sitting at a big CRT screen with all kinds of symbols, new and different symbols, not restricted to our old ones. The computer could be manipulated, and you could be operating all kinds of things to drive the computer

... I also had a clear picture that one's colleagues could be sitting in other rooms with similar work stations, tied to the same computer complex, and could be sharing and working and collaborating very closely. And also the assumption that there'd be a lot of new skills, new ways of thinking that would evolve"

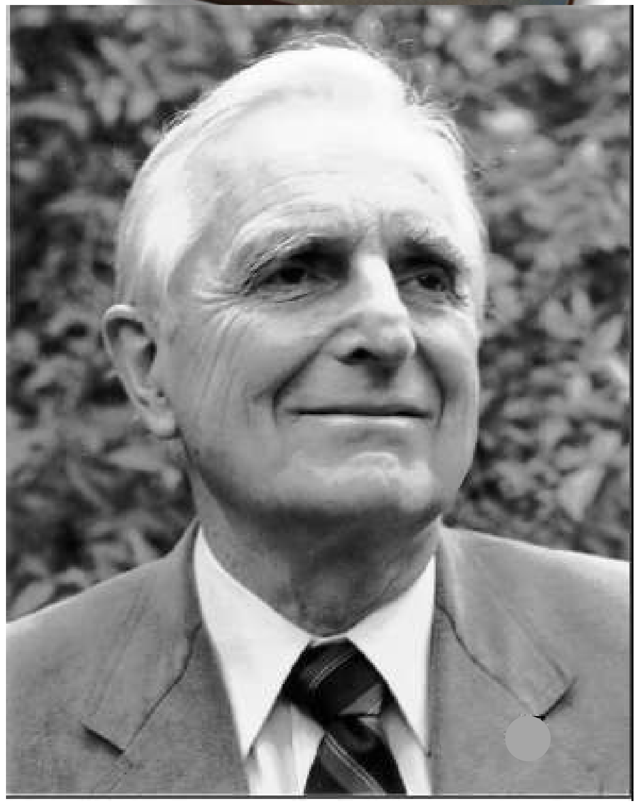


History of HCI

A Conceptual Framework for Augmenting Human Intellect (SRI Report, 1962)
http://www.1962paper.org/web.html

"By *augmenting man's intellect* we mean increasing the capability of a man to approach a complex problem situation, gain comprehension to suit his particular needs, and to derive solutions to problems.

One objective is to develop new techniques, procedures, and systems that will better adapt people's basic information-handling capabilities to the needs, problems, and progress of society."



History of HCI

A Conceptual Framework for Augmenti
Human Intellect (SRI Report, 19
http://www.1962paper.org/web.html

"By *augmentin*
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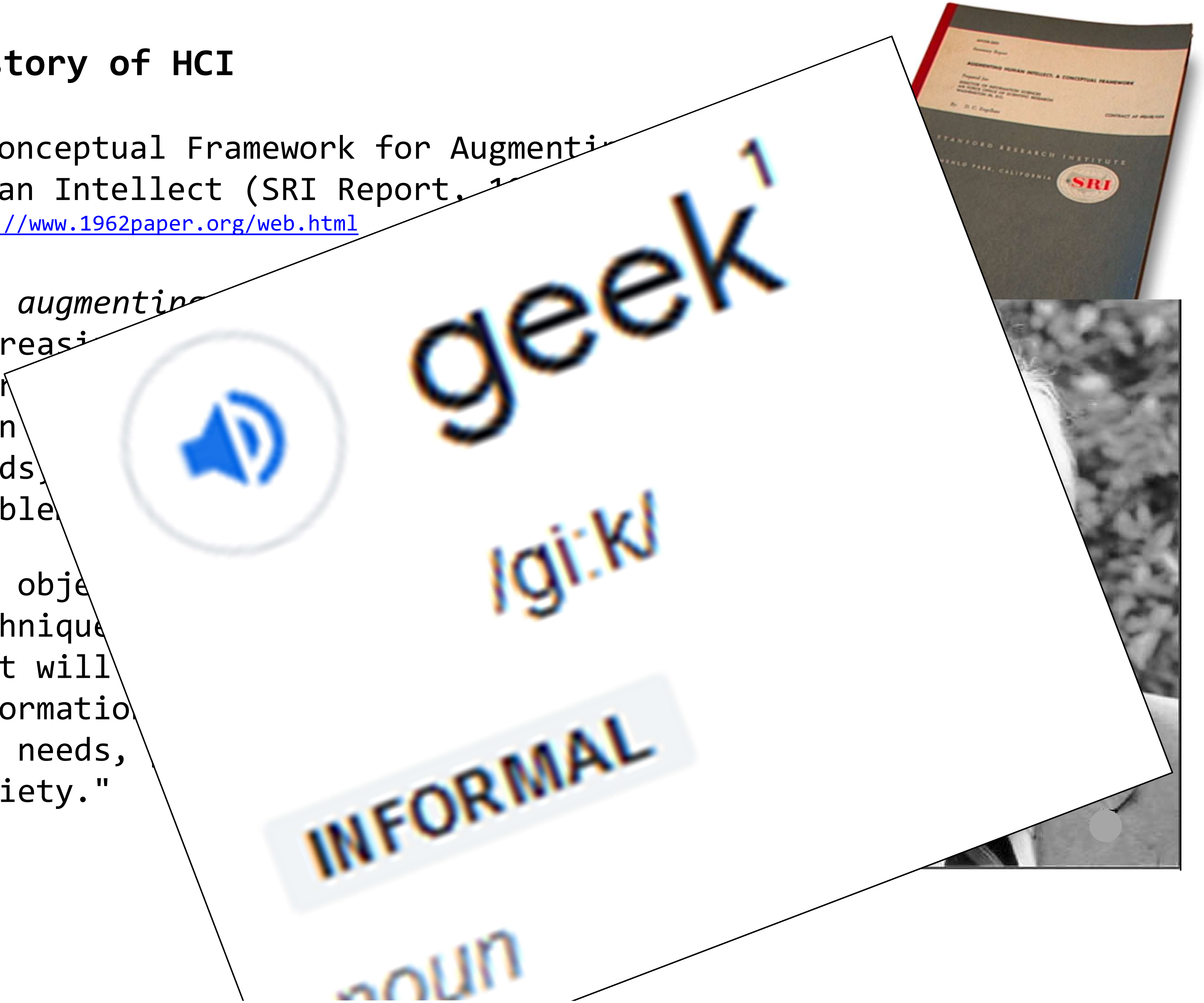
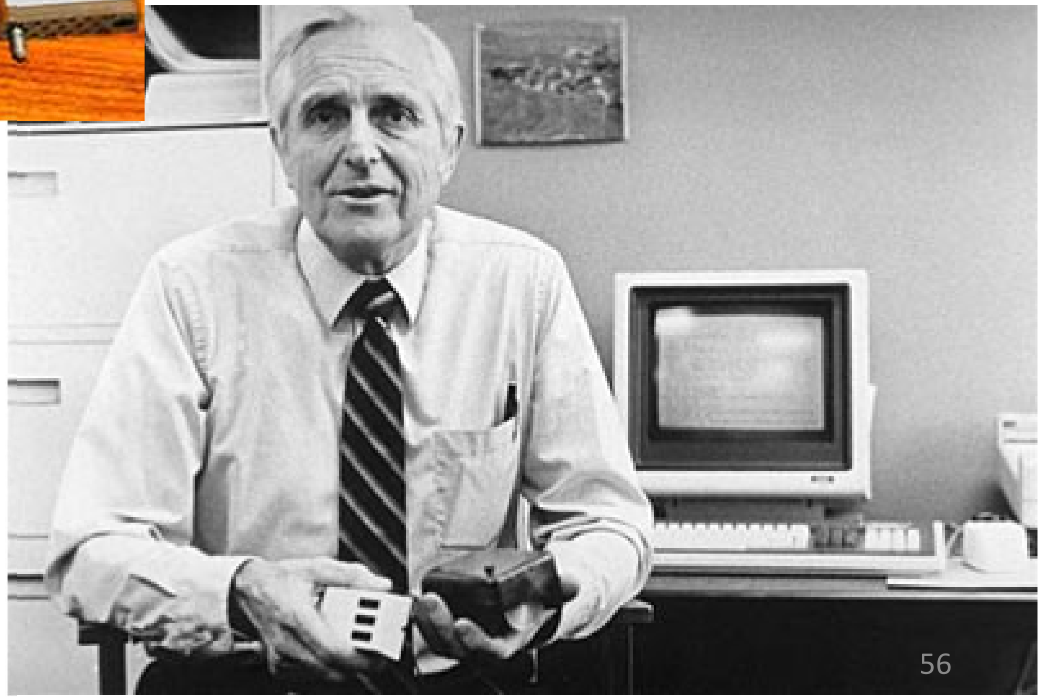


History of HCI

Douglas Engelbart (1964)

First mouse

Gamers, pay tribute!



HCI Summary

That’s it for today:

HCI leads to cost-reduction (less redesign)
HCI is what sells (why else buy Apple?)
Avoids disasters (financial, airospace, nuclear, self-driving automotive)

History: from little interaction and human subordinate to the system,
to much interaction and system tending to human needs

Visionaries come up with next-generation modes of interaction

Cherish your geeks!